

# action-translation

**AI-Powered Translation Automation for Lectures**

GitHub Action + `translate` CLI • Claude Sonnet 5 • MyST Markdown

*QuantEcon Project*

# What It Does

**Automatically translates and reviews lectures when source content changes**

Monitor merged PRs → Detect changes → Translate → Create PR → AI Review → Route

## Key Capabilities

- ✓ **Smart Diff Translation** – Only translates modified sections
- ✓ **MyST Markdown Aware** – Preserves code, math, directives, anchors
- ✓ **Consistent Terminology** – Per-language glossaries (330–370 terms each)
- ✓ **Machine-Readable Review** – Verdict block with scores + auto-merge/editor routing
- ✓ **Deterministic Guardrails** – Structural parity, typography, bibliography backfill
- ✓ **Language Extensible** – zh-cn, fa, fr, ml, ja, es configured

# How It Works

## Section-Based Translation Approach

### Problem: Block-Level

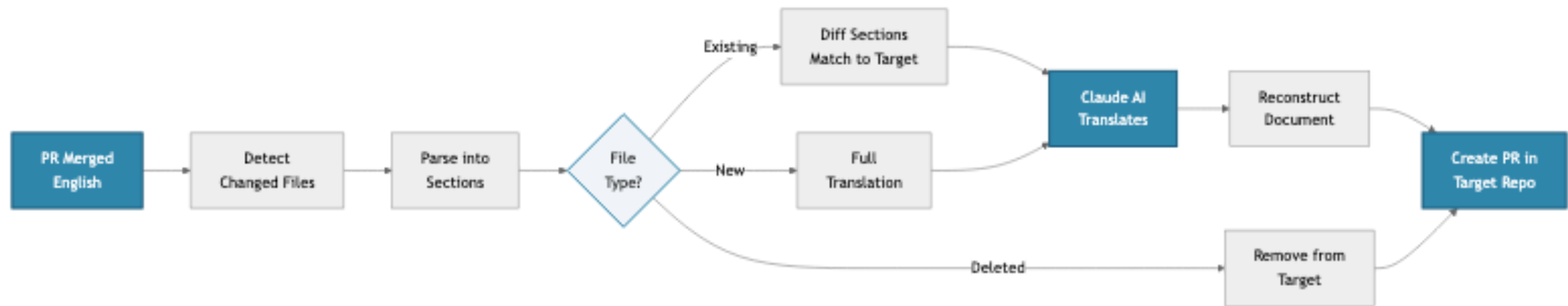
- Can't match across languages
- Loses translation context
- Complex mapping logic

### Solution: Section-Based

- Match by heading-map, then position
- Translate full sections
- Simple: Add, Update, Delete

Heading maps live in each translated file's frontmatter — English heading text mapped to its translation, so sections stay matched even after edits and reorders.

# Translation Workflow



# LLM-Powered Translation (Claude Sonnet 5)

## UPDATE Mode

*(changed sections)*

- Sends: old EN + new EN + current translation
- Claude sees what changed
- Preserves style, terminology, and target-side localisation
- Glossary pins technical terms

## NEW / RESYNC Modes

*(new files, drift recovery)*

- Whole-document translation with full context
- Deterministic post-pass: typography, heading map, frontmatter carry-forward
- Structural parity guard refuses corrupted output

# Three Operational Modes

## Sync Mode (*source repo*)

- Fires on merged PRs
- Translates changed sections
- Backfills cited `.bib` entries
- One PR per target language

## Rebase Mode (*target repo*)

- Keeps sibling translation PRs current during waves

## Review Mode (*target repo*)

- Scores accuracy, fluency, terminology, formatting
- Deterministic diff checks (structure, heading map)
- Emits a machine-readable verdict block
- Routes: `auto-merge` vs `editor`
- Shadow mode records would-auto-merge without acting

# The `translate` CLI

Operator tooling for everything the Action can't reach from CI

| Command                                  | Purpose                                                            |
|------------------------------------------|--------------------------------------------------------------------|
| <code>init</code>                        | Bulk-seed a new edition from the source repo                       |
| <code>status</code>                      | Per-file sync state ( <code>--check-sync</code> for content drift) |
| <code>forward</code>                     | Resync drifted files, one PR each ( <code>--github</code> )        |
| <code>backward</code>                    | Capture target-side improvements upstream                          |
| <code>review</code>                      | Local review of a translation PR                                   |
| <code>setup</code> / <code>doctor</code> | Scaffold and verify workflow wiring                                |

# Trust & Safety

- ✓ **Trust-gated triggers** – resync commands require OWNER/MEMBER/COLLABORATOR
- ✓ **Fail-closed parsing** – malformed model output fails the run, never fabricates a verdict
- ✓ **Verdict provenance** – engine version + reviewed SHA in every verdict block
- ✓ **Append-only bibliography** – localised entries are never overwritten
- ✓ **Loud failure** – partial runs open issues; silent-success paths are being closed (W1)



# Status & Getting Started

## Current Status

- 📦 **v0.25.0** – deployed estate-wide
- 🔄 **Three modes:** Sync + Review + Rebase
- ✅ 1,500+ tests, type-checked, zero skips
- 🔧 27 E2E scenarios × 3 languages on real repos

## In Production

- lecture-python.zh-cn
- lecture-python-programming .zh-cn / .fa / .fr

## Resources

**GitHub:** QuantEcon/action-translation

**Docs:** `docs/user/` (quickstart, tutorials, metadata contract)

**License:** MIT